Type MS · Sketching curves and where they cross the axes

Answer 1 of 11 · Question 3 part e · 2 marks

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3 a	$8\left(x^2+\frac{10}{8}x\right)-3$	M1
	$8\left(x+\frac{5}{8}\right)^2-\frac{200}{64}-3=8\left(x+\frac{5}{8}\right)^2-\frac{49}{8}$	M1
	So $A=8$ $B=\frac{5}{8}$ $C=-\frac{49}{8}$	A1 (3)
b (i)	$x=-\frac{5}{8}$	B1 ft
(ii)	$f(x)_{min}=-\frac{49}{8}$	B1 ft (2)
c	$8\left(x+\frac{5}{8}\right)^2-\frac{49}{8}=0\Rightarrow x=-\frac{3}{2}$ $x=\frac{1}{4}$	M1 A1 (2)
d	$8x^2+10x-3=2x+13\Rightarrow 8x^2+8x-16=0$ oe	M1
	$x^2+x-2=0=(x+2)(x-1)\Rightarrow x=-2,\ x=1$	M1 A1
	$y=9$ $y=15$ Coordinates are $(-2,9)$ and $(1,15)$	A1 (4)
e	Correct curve or line Correct curve and line	B1 B1 (2)
		Total 13 marks

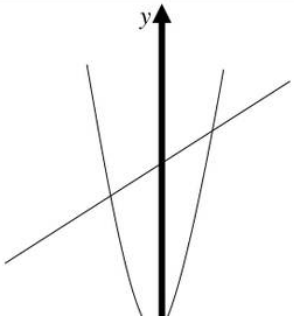
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(b) (i)	B1ft	For $x=-\frac{5}{8}$ ft their $-B$ Allow differentiation: $\frac{dy}{dx}=16x+10=0\Rightarrow x=-\frac{10}{16}$ which must be correct.
(ii)	B1ft	For $f(x)_{min}=-\frac{49}{8}$ ft their C If they differentiate, allow a ft from their differentiation.
	Incorrect or no labelling of parts. If the responses are labelled incorrectly, mark as labelled. If there is no labelling, treat the first as (i) and the second as (ii) and mark accordingly.	
(c)	M1	For setting the given f(x) = 0 and solving using any method. See General Guidance.
	A1	For $x=-\frac{3}{2}$ and $x=\frac{1}{4}$
(d)	M1	For setting $8x^2+10x-3=2x+13$ and forming a 3TQ which as a minimum must be $8x^2+8x\pm C$ where C is a constant, or any simplification, e.g., $x^2+x\pm K$ where K is a constant.
	M1	For attempting to solve their 3TQ. See General Guidance.
	A1	For **both** $x=-2$, $x=1$
	A1	For $(-2,9)$ and $(1,15)$ Accept $x=-2$, $y=9$ and $x=1$, $y=15$ paired correctly.
	ALT – solves the equation in y	
	M1	Substitutes $x=\frac{y-13}{2}$ into $y=8x^2+10x-3$ and forms a 3TQ which as a minimum must be of the form $2y^2-36y\pm X$ $X\neq 0$ $\Rightarrow y=8\left(\frac{y-13}{2}\right)^2+10\left(\frac{y-13}{2}\right)-3\Rightarrow 2y^2-36y+75=0$
	M1	For attempting to solve their 3TQ. See General Guidance.
	A1	For **both** $y=9$ and $y=15$
	A1	For $(-2,9)$ and $(1,15)$ Accept $x=-2$, $y=9$ and $x=1$, $y=15$ paired correctly.
(e)	B1	For correct curve (intersections with x-axis are at $(-1.5, 0)$ and $(0.25, 0)$) **or** line (Intersections are $(-6.5, 0)$ and $(0, 13)$ drawn. **We do not need to see any of these points marked. These are for guidance only.** The question asks for a sketch, accept a reasonable attempt. • Accept a positive quadratic curve with the minimum point below the x-axis, and one branch either side of the y-axis. • Accept a straight line with a positive gradient where the intersection with the y-axis is positive.
	B1	For correct curve **and** line drawn

SKETCH OF CURVE AND LINE



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Question number	Scheme	Marks
4	When $x=\frac{\pi}{2}$ $y=\frac{\pi^3}{8}$ So $\left(\frac{\pi}{2},\frac{\pi^3}{8}\right)$	B1
	$\frac{dy}{dx}=3x^2\sin x+x^3\cos x$	M1 A1 A1
	When $x=\frac{\pi}{2}$ $\frac{dy}{dx}=3\left(\frac{\pi}{2}\right)^2\sin\left(\frac{\pi}{2}\right)+\left(\frac{\pi}{2}\right)^3\cos\left(\frac{\pi}{2}\right)=\left[,\frac{3\pi^2}{4},\right]$	M1
	$y-\frac{\pi^3}{8}=\frac{3\pi^2}{4}\left(x-\frac{\pi}{2}\right)$	M1
	$y=\frac{3}{4}\pi^2x-\frac{1}{4}\pi^3$	A1
		Total 7 marks

Type MS · Sketching curves and where they cross the axes

Answer 1 of 11 · Question 3 part e · 2 marks

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Mark	Notes
Note: In this question, all substitution of angle values must be in Radians only.	
B1	For obtaining $y = \frac{\pi^3}{8}$ [allow $\left(\frac{\pi}{2}\right)^3$ and also awrt $y = 3.88$]
M1	For an attempt to use the product rule. The definition of an attempt is as follows: • There must be a correct attempt to differentiate both terms. $\sin x \Rightarrow \cos x \quad x^3 \Rightarrow ax^2$ where $x \neq 0$ • The correct formula must be used. i.e., it must be a sum of their two terms.
A1	At least one term must be correct. Either $3x^2 \sin x$ or $x^3 \cos x$
A1	For $3x^2 \sin x + x^3 \cos x$ oe Ignore any subsequent simplification once you have seen the correct answer – even if the simplification is incorrect.
M1	For substitution of $\frac{\pi}{2}$ into their $\frac{dy}{dx} = \left['\frac{3}{4}\pi^2'\right]$ provided it is a changed expression. Allow a value of awrt 7.4(0) **NOTE:** You must see a full substitution of $\frac{\pi}{2}$ into their $\frac{dy}{dx}$ if their expression for $\frac{dy}{dx}$ is incorrect.
M1	For a correct method for finding the equation of a line using their value of y, dy/dx and the given x [allow $x = 1.57\ldots$]. This must be applied correctly. Either uses the formula with their values, or if uses $y = mx + c$ they must reach a value for c before this mark can be awarded. Do **not** allow processing errors to find the value of c
A1	For $y = \frac{3}{4}\pi^2 x - \frac{1}{4}\pi^3$ Allow $y = 7.4(0)x - 7.75$ or better values. Do not allow a mixture of decimals and exact values.

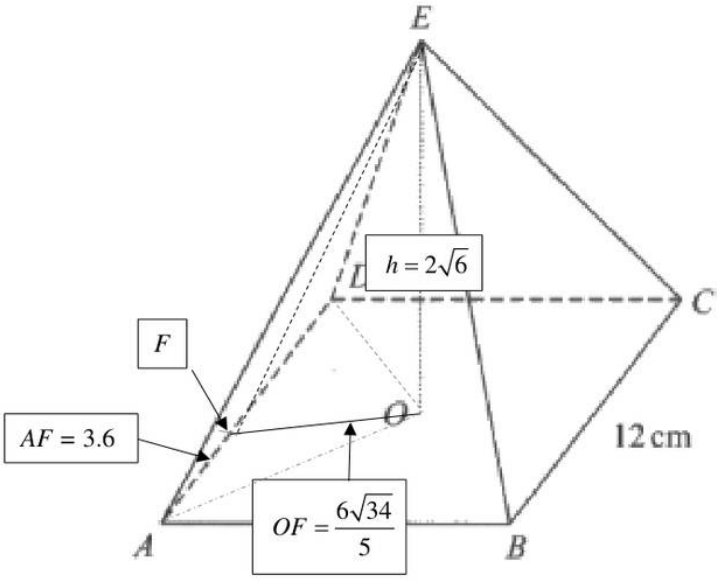
Type MS · Sketching curves and where they cross the axes

Answer 1 of 11 · Question 3 part e · 2 marks

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Question number	Scheme	Marks
5 a	$OC = \frac{1}{2}\sqrt{12^2+12^2} = \left[6\sqrt{2}\right]$ or $AC = \sqrt{12^2+12^2} = \left[12\sqrt{2}\right]$	M1
	$h = EO = 6\sqrt{2} \times \frac{\sqrt{3}}{3} = 2\sqrt{6}$ oe	M1 A1
		(3)
b	Midpoint of AD to $F = \left(6 - \frac{12}{1+4}\right)$ [= 3.6]	M1
	$OF = \sqrt{6^2+3.6^2} = \frac{6\sqrt{34}}{5}$	M1
	$\tan\theta = \frac{2\sqrt{6}}{\frac{6\sqrt{34}}{5}} = 35^\circ$	M1 A1
		(4)
		Total 7 marks

USEFUL SKETCH



Type MS · Sketching curves and where they cross the axes

Answer 1 of 11 · Question 3 part e · 2 marks

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Part	Mark	Notes
(a)	M1	For using a correct Pythagoras theorem to find either OC, OA $\left(6\sqrt{2}\right)$ or AC $\left(12\sqrt{2}\right)$
	M1	For using the correct trigonometry (tan ratio): $h = '6\sqrt{2}'\tan 30^\circ = \left[2\sqrt{6}\right]$ or equivalent. Eg $\tan 30 = \dfrac{h}{6\sqrt{2}} \Rightarrow h = ...$
	A1	For $2\sqrt{6}$ oe
(b)	M1	For correct expression for the midpoint of AD to F $\left(6 - \dfrac{12}{1+4}\right)$ or 3.6 oe seen
	M1	For the correct use of Pythagoras' to find OF $\dfrac{6\sqrt{34}}{5}$ oe [6.997..] ft their 3.6
	ALT	
	M1	Finds the length OF using cosine rule. $OF = \sqrt{2.4^2 + \left(6\sqrt{2}\right)^2 - 2\times 2.4\times 6\sqrt{2}\times\cos 45} = \left[\dfrac{6\sqrt{34}}{5}\right]$ OR $OF = \sqrt{9.6^2 + \left(6\sqrt{2}\right)^2 - 2\times 9.6\times 6\sqrt{2}\times\cos 45} = \left[\dfrac{6\sqrt{34}}{5}\right]$ OR $\cos 45^\circ = \dfrac{9.6^2 + \left(6\sqrt{2}\right)^2 - OF^2}{2\times 9.6\times 6\sqrt{2}} \Rightarrow OF = ...$ oe [6.997..]
	M1	For the correct evaluation of their cosine rule. $\left(\dfrac{6\sqrt{34}}{5}\text{ oe}\right)$ Accept awrt 7.00 [6.997142…]
	dM1	For the correct use of $\tan\theta = \dfrac{'EO'}{'OF'}$ This mark is dependent on both previous M marks. They must have a valid method to find OF for the award of this mark.
	A1	For 35(°) or better (Calculator value is 34.997…°.)
	Note: There are other methods – if unsure, send to Review.	

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Answer 1 of 11 · Question 3 part e · 2 marks

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Question number	Scheme	Marks
6 a	$r=\frac{q(2p-3)}{q(2p+3)}=\frac{q(2p+3)}{q(4p+1)}$	M1
	$(4p+1)(2p-3)=(2p+3)^2$	dM1
	$2p^2-11p-6=0$	A1
	$(2p+1)(p-6)=0$	M1
	$p=-\frac{1}{2}$ or $p=6$	A1 (5)
b	When $p=6$ $r=\frac{3}{5}$ and $U_1=q\left(4\times'6'+1\right)=\left(25q\right)$	M1
	$S_\infty=\frac{25q}{\frac{2}{5}}=250\Rightarrow q=4$	dM1 A1 (3)
		Total 8 marks

Type MS · Sketching curves and where they cross the axes

Answer 1 of 11 · Question 3 part e · 2 marks

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Part	Mark	Notes
(a)	M1	For $r = \frac{\text{3rd term}}{\text{2nd term}} = \frac{\text{2nd term}}{\text{1st term}}$
	dM1	For attempting to remove the denominators/simplifying the expression to attempt to obtain a 3TQ. Allow one processing error. Allow a minimally acceptable $4p^2 \pm Xp \pm Y = (0)$ or $2p^2 \pm Pp \pm Q = (0)$ $X, Y, P, Q \neq 0$ **Note:** This mark is dependent on the previous M mark.
	A1	For obtaining $2p^2 - 11p - 6 = (0)$ or equivalent. Their working will give them $4p^2 - 22p - 12 = (0)$ before further simplification
	M1	For a valid attempt to solve the 3TQ. Note this is an independent mark for their 3TQ. See General Guidance. The 3TQ must have come from some attempted manipulation involving $q(2p-3)$, $q(2p+3)$ and $q(4p+1)$
	A1	For $p = -\frac{1}{2}$ or $p = 6$
(b)	M1	For finding: • A value of r for their p. Allow any value of r $r \neq 0$ E.g. $r = \frac{(2\times'6'-3)}{(2\times'6'+3)} = ...$ $\left(\text{Note: for } p = -\frac{1}{2} \quad r = -2\right)$ • A value for U_1 $U_1 = q(4\times'6'+1) = ...$ $\left(\text{Note: for } p = -\frac{1}{2}, U_1 = -q\right)$ You may see a = 100 after later working to find the sum to infinity. If their values of p are incorrect, we **must** see working here. **NOTE:** This is a B mark in Epen.
	dM1	For use of $S_\infty = \frac{a}{1-r} = 250$ with their r and their U_1 provided $\lvert r\rvert < 1$ If they use the formula for the sum to infinity on an $\lvert r\rvert > 1$ withhold this mark even if they use it on one valid and one invalid attempt. **Note:** This mark is dependent on the previous M mark.
	A1	For $q = 4$

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Answer 1 of 11 · Question 3 part e · 2 marks

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Question number	Scheme	Marks
7 a	$\frac{dy}{dx} = 2e^{2x}\cos 2x - 2e^{2x}\sin 2x$	M1 A1 A1
	$\frac{dy}{dx} = 2y - 2e^{2x}\sin 2x$ *	A1 cso (4)
b	$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$	M1 A1
	$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} + 2\left(\frac{dy}{dx} - 2y\right) - 4y$	dM1ddM1
	$\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$ *	A1 cso (5)
		Total 9 marks

Part	Mark	Notes
(a)	M1	For an attempt to use the product rule. The definition of an attempt is as follows: • There must be a correct attempt to differentiate both terms. $\cos 2x \Rightarrow \pm 2\sin 2x \quad e^{2x} \Rightarrow 2e^{2x}$ where $x \neq 0$ • The terms must be added The correct formula must be used.
	A1	For one term correct
	A1	For both terms correct
	A1 cso	For obtaining the given result with no errors seen.
(b)	**Method 1**	
	M1	For attempting to differentiate $\frac{dy}{dx}$ to obtain $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 4e^{2x}\cos 2x - 4e^{2x}\sin 2x - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$ For this mark accept either: $\left(4e^{2x}\cos 2x - 4e^{2x}\sin 2x\right)$ **OR** $-4e^{2x}\sin 2x - 4e^{2x}\cos 2x$
	A1	For the correct $\frac{d^2y}{dx^2}$ (This is an M mark in Epen)
	dM1	For simplifying $\frac{d^2y}{dx^2}$ to obtain $\frac{d^2y}{dx^2} = -8e^{2x}\sin 2x$ (A mark in Epen) **This mark is dependent on the first M mark**

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Answer 1 of 11 · Question 3 part e · 2 marks

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	ddM1	For using the substitution $-2e^{2x}\sin 2x = \frac{dy}{dx} - 2y \Rightarrow -8e^{2x}\sin 2x = 4\frac{dy}{dx} - 8y$ **This mark is dependent on BOTH previous M marks.**
	A1cso	For obtaining the given result with no errors seen. $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$
	Method 2	
	M1	For attempting to differentiate $\frac{dy}{dx}$ to obtain $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 2\frac{dy}{dx} - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$ For this mark accept either: $2\frac{dy}{dx}$ or $-4e^{2x}\sin 2x - 4e^{2x}\cos 2x$
	A1	For the correct $\frac{d^2y}{dx^2}$ **This is an M mark in Epen**
	dM1	For substituting $-4e^{2x}\cos 2x \Rightarrow -2y$ **This is an A mark in Epen**
	ddM1	For using the substitution $-4e^{2x}\sin 2x = 2\frac{dy}{dx} - 4y$
	A1	For obtaining the given result with no errors seen. $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$
	Method 3 – Works form LHS and RHS together.	
	M1	For attempting to differentiate $\frac{dy}{dx}$ to obtain $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 4e^{2x}\cos 2x - 4e^{2x}\sin 2x - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$ For this mark accept either: $\left(4e^{2x}\cos 2x - 4e^{2x}\sin 2x\right)$ **OR** $-4e^{2x}\sin 2x - 4e^{2x}\cos 2x$
	A1	For the correct $\frac{d^2y}{dx^2}$ **This is an M mark in Epen**
	dM1	For simplifying $\frac{d^2y}{dx^2}$ to obtain $\frac{d^2y}{dx^2} = -8e^{2x}\sin 2x$ (A mark in Epen) **This mark is dependent on the first M mark**
	ddM1	Multiplies out the given expression in (b) $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y = 4\left(2e^{2x}\cos 2x - 2e^{2x}\sin 2x\right) - 8\left(e^{2x}\cos 2x\right)$ $= 8e^{2x}\cos 2x - 8e^{2x}\sin 2x - 8e^{2x}\cos 2x$ $= -8e^{2x}\sin 2x$

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		This mark is dependent on BOTH previous M marks.
	A1	For a conclusion. A simple # sign, 'shown', 'QED' or underlining is sufficient.

Question number	Scheme	Marks
8	$\alpha+\beta=4k\sqrt{2}$ and $\alpha\beta=2k^4-1$	B1 B1
	$(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta\Rightarrow\left(4k\sqrt{2}\right)^2=66+2\left(2k^4-1\right)$	M1 A1
	$k^4-8k^2+16=0$	M1
	$\left(k^2-4\right)^2=0\Rightarrow k=2$	M1 A1
	$(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3$ **or** $\alpha^3+\beta^3=(\alpha+\beta)\left(\alpha^2+\beta^2-\alpha\beta\right)$	M1
	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ **or** $\alpha^3+\beta^3=(\alpha+\beta)(66-\alpha\beta)$	A1
	$\alpha^3+\beta^3=\left(8\sqrt{2}\right)^3-3\times31\times8\sqrt{2}=280\sqrt{2}$ **or** $8\sqrt{2}(66-31)=280\sqrt{2}$	M1
	p=280	A1 (11)
		Total 11 marks

Mark	Notes
B1	For **either** $\alpha+\beta=4k\sqrt{2}$ **or** $\alpha\beta=2k^4-1$
B1	For **both** $\alpha+\beta=4k\sqrt{2}$ **and** $\alpha\beta=2k^4-1$
M1	For the correct algebra on $\alpha^2+\beta^2$ (in any order) **and** substitution of their values of $\alpha\beta$ and $\alpha+\beta$ providing both sum and product are in terms of k. $(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta\Rightarrow\alpha^2+\beta^2=\left('4k\sqrt{2}'\right)^2-2\left('2k^4-1'\right)$
A1	For obtaining $\left(4k\sqrt{2}\right)^2=66+2\left(2k^4-1\right)$ in any order.
M1	For simplifying to form a 3TQ in k^4 i.e., $4k^4-32k^2+64=0$ oe Accept as a minimum $4k^4-32k^2\pm Q=(0)\quad Q\neq0$
M1	For factorising or solving the 3TQ using any valid method. See General Guidance.
A1	For $k=2$ If they also give $k=-2$ withhold this mark.

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Method 1	
M1	For expanding $(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3$ or $\alpha^3+\beta^3=(\alpha+\beta)(\alpha^2+\beta^2-\alpha\beta)$
A1	For obtaining $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ or $\alpha^3+\beta^3=(\alpha+\beta)(66-\alpha\beta)$ This must be such that $\alpha\beta$ and $\alpha+\beta$ can be substituted in directly.
M1	For substitution of $\alpha+\beta$ and $\alpha\beta$ for their positive value of k into a **correct** expansion of $\alpha^3+\beta^3$ **NOTE:** If they do not obtain $k=2$, then full substitution of their numerical value for k into $\alpha+\beta$ **and** $\alpha\beta$ must be seen for the award of this mark. For example: $\alpha^3+\beta^3=\left(4\times\text{'their } k\text{'}\sqrt{2}\right)^3-3\times\left(2\times\text{'their } k^4\text{'}-1\right)\times\left(4\times\text{'their } k\text{'}\sqrt{2}\right)$
A1	For $p=280$
Method 2	
M1	Finds the exact value of α and β Solves the equations $\left(\alpha\beta=2k^4-1 \text{ and } \alpha+\beta=4k\sqrt{2}\right)$ simultaneously to give a value for α and β $\alpha=\frac{31}{\beta}\Rightarrow\alpha+\beta=\frac{31}{\beta}+\beta=8\sqrt{2}\Rightarrow\beta^2-8\sqrt{2}\beta+31=0$ $\Rightarrow\beta=\ldots\quad\alpha=\ldots$
A1	For $\alpha=1+4\sqrt{2}\quad\beta=-1+4\sqrt{2}$ OR $\beta=1+4\sqrt{2}\quad\alpha=-1+4\sqrt{2}$
M1	Substitutes these values into $\alpha^3+\beta^3=p\sqrt{2}$ to find a value for p $\left(1+4\sqrt{2}\right)^3+\left(-1+4\sqrt{2}\right)^3=24\sqrt{2}+256\sqrt{2}=280\sqrt{2}\Rightarrow p=\ldots$
A1	$p=280$

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Question number	Scheme	Marks
9		
	$\frac{dA}{dt} = 0.45$	B1
	$V = x^3 \Rightarrow \frac{dV}{dx} = 3x^2$	B1
	$A = 6x^2 \Rightarrow \frac{dA}{dx} = 12x$	B1
	$384 = 6x^2 \Rightarrow x = 8$	M1
	$\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dA} \times \frac{dA}{dt}$	M1
	$\frac{dV}{dt} = 3x^2 \times \frac{1}{12x} \times 0.45 \left[= \frac{9x}{80} \right]$ oe	dM1
	When $x = 8$ $\frac{dV}{dt} = 0.9\,\text{cm}^3/\text{s}$	A1
		Total 7 marks

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Mark	Notes
B1	For $\frac{dA}{dt} = 0.45$ seen anywhere in their working Accept other letters, for example S for the area $\frac{dS}{dt} = 0.45$
B1	For $\frac{dV}{dx} = 3x^2$ Accept also other letters in place of x such as r for example.
B1	For $\frac{dA}{dx} = 12x$ Accept also other letters in place of x such as r for example.
M1	For setting $384 = 6x^2$ and proceeding to a correct method leading to a value of x Award this mark when they obtain $x^2 = 64 \Rightarrow x = ...$
M1	For a correct expression of the chain rule seen or implied. i.e., $\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dA} \times \frac{dA}{dt}$ They may complete this in two stages. So you may see for example: $\frac{dx}{dt} = \frac{1}{\frac{dA}{dx}} \times \frac{dA}{dt}$ **AND** $\frac{dV}{dt} = \frac{dx}{dt} \times \frac{dV}{dx}$
dM1	For substituting their values into a **correct** chain rule. $\frac{dV}{dt} = 3(8)^2 \times \frac{1}{12(8)} \times 0.45$ This mark is dependent on the previous M mark scored.
A1	For 0.9 (cm^3/s)

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Answer 1 of 11 · Question 3 part e · 2 marks

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Question number	Scheme	Marks
10 a (i)	$\sin 2\theta = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$ *	M1 A1 cso (2)
a (ii)	$\cos 2\theta = \cos^2\theta - \sin^2\theta = \cos^2\theta - (1-\cos^2\theta)$	M1 M1
	$= 2\cos^2\theta - 1$ *	A1 cso (3)
b	$\sin 2\theta - \tan\theta = 2\sin\theta\cos\theta - \dfrac{\sin\theta}{\cos\theta}$	M1
	$\sin\theta\left(2\cos\theta - \dfrac{1}{\cos\theta}\right)$	M1
	$\sin\theta\left(\dfrac{2\cos^2\theta - 1}{\cos\theta}\right) = \tan\theta\cos 2\theta$ *	dM1 A1cso (4)
c	$\tan x = 0$ so $x = 180$	B1
	$\cos 2x = 0$ so $2x = 90, 270, 450, 630$	M1
	$x = 45, 135, 225, 315$	A1 A1 (4)
		Total 13 marks

Part	Mark	Notes
(a)(i)	M1	For the correct use of: $\sin(A+B) = \sin A\cos B + \cos A\sin B \Rightarrow \sin 2\theta = \sin\theta\cos\theta + \sin\theta\cos\theta$ Allow any letter to be used for this mark.
	A1 Cso	For obtaining the given result $\sin 2\theta = 2\sin\theta\cos\theta$ or allow $\sin(\theta+\theta) = 2\sin\theta\cos\theta$
(a)(ii)	M1	For the correct use of: $\cos(A+B) = \cos A\cos B - \sin A\sin B$ $\Rightarrow \cos 2A = \cos^2 A - \sin^2 A$ or $\cos(A+A) = \cos A\cos A - \sin A\sin A$ Allow any letter to be used for this mark.
	M1	For use of $\sin^2 A + \cos^2 A = 1$ and substituting Allow any letter to be used for this mark.
	A1 cso	For obtaining the given result $\cos 2\theta = 2\cos^2\theta - 1$ or allow $\cos(\theta+\theta) = 2\cos^2\theta - 1$
(b)	M1	For correct use of $\sin 2\theta = 2\sin\theta\cos\theta$ **and** $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$

Type MS · Sketching curves and where they cross the axes

Answer 1 of 11 · Question 3 part e · 2 marks

2023 JAN · Paper 02 · extract 16 of 20 · find part e in this mark scheme extract

	M1	For factorising the resulting expression to give $\sin\theta\left(2\cos\theta-\frac{1}{\cos\theta}\right)$ **and** using a common denominator of $\cos\theta$ correctly to obtain $\sin\theta\left(\frac{2\cos^2\theta-1}{\cos\theta}\right)$ oe e.g. $\frac{\sin\theta}{\cos\theta}\left(2\cos^2\theta-1\right)$
	dM1	For use of $2\cos^2\theta-1=\cos 2\theta$ **Note:** This mark is dependent on both previous M marks
	A1 cso	For obtaining the given result
	ALT - Allow candidates to work from both sides. For example.	
	M1	For correct use of $\sin 2\theta=2\sin\theta\cos\theta$ **and** $\tan\theta=\frac{\sin\theta}{\cos\theta}$ $\sin 2\theta-\tan\theta=\tan\theta\cos 2\theta$ $\Rightarrow 2\sin\theta\cos\theta-\frac{\sin\theta}{\cos\theta}=\frac{\sin\theta}{\cos\theta}\left(2\cos^2\theta-1\right)$
	M1	For using a common denominator on the LHS to give: $\frac{2\sin\theta\cos^2\theta-\sin\theta}{\cos\theta}=\frac{\sin\theta\left(2\cos^2\theta-1\right)}{\cos\theta}$
	dM1	Multiplies or factorises one side to make LHS = RHS For example; $\frac{2\sin\theta\left(\cos^2\theta-1\right)}{\cos\theta}=\frac{2\sin\theta\left(\cos^2\theta-1\right)}{\cos\theta}$
	A1 cso	Concludes RHS = LHS therefore shown. Note there must be no errors seen. If unsure, send to Review.
(c)	Some candidates are starting again without using the given expression in (b) and are getting their angles from cos x rather than cos $2x$ Allow the use of θ or any other variable throughout this part of the question. As a general principle of marking this part: • B1 – For only 180° seen. If 0° is included, then B0. • M1 – For either $2x=90,(270,450,630)$ **OR** $x=45^\circ$ or 135° or 225° or 315° or any other correct angle in or out of range. • A1 – For any two correct angles apart from 180° • A1 – For all four correct angles, apart form 180°	
	B1	For $x=180$ If $x=0$ included then B0
	M1	For obtaining $2x=90,(270,450,630)$
	A1	For any two correct angles from 45°, 135°, 225° or 315°
	A1	For all four angles from $\cos x$ or $\cos 2x$
	For extra angles within range, penalise the last A mark only. For example; If they have [coming from a correct method] $x=45°, 67.5°, 135°, 225°, 315°$ score M1A1A0 If they have $x=45°, 67.5°, 135°$ then award M1A1A0	

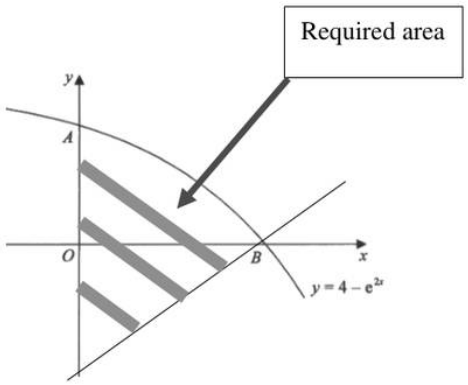
Type MS · Sketching curves and where they cross the axes

Answer 1 of 11 · Question 3 part e · 2 marks

2023 JAN · Paper 02 · extract 17 of 20 · find part e in this mark scheme extract

Question number	Scheme	Marks
11 a	When $x=0 \quad y=3$	B1
	When $y=0 \quad 0=4-e^{2x} \Rightarrow \ln 4 = 2x \Rightarrow x = \ln 2$	M1 A1 (3)
b	$\frac{dy}{dx} = -2e^{2x}$	M1
	When $x=\ln 2 \quad \frac{dy}{dx} = -8$	M1
	$y-0=\frac{1}{8}(x-\ln 2) \Rightarrow y=\frac{1}{8}x-\frac{1}{8}\ln 2$	M1 A1 (4)
c	$\int_0^{\ln 2}\left(4-e^{2x}\right)dx=\left[4x-\frac{1}{2}e^{2x}\right]_0^{\ln 2}=\left(4\ln 2-\frac{1}{2}e^{2\ln 2}\right)-\left(0-\frac{1}{2}e^{0}\right)$	M1M1M1
	$\frac{1}{2}(\ln 2)\left(\frac{1}{8}\ln 2\right)=\frac{1}{16}(\ln 2)^2$	M1M1
	$\frac{1}{8}\int_0^{\ln 2}\lvert x-\ln 2\rvert\,dx=\frac{1}{8}\left\lvert\frac{x^2}{2}-x\ln 2\right\rvert_0^{\ln 2}=\left\lvert\frac{(\ln 2)^2}{16}-\frac{(\ln 2)^2}{8}\right\rvert=\frac{(\ln 2)^2}{16}$	[M1M1]
	$4\ln 2-\frac{3}{2}+\frac{1}{16}(\ln 2)^2=1.3$	M1 A1 (7)
		Total 14 marks

USEFUL SKETCH



Part	Mark	Notes

Type MS · Sketching curves and where they cross the axes

Answer 1 of 11 · Question 3 part e · 2 marks

2023 JAN · Paper 02 · extract 18 of 20 · find part e in this mark scheme extract

(a)(i) (ii)	B1	For $y = 3$
	M1	For $\ln 4 = 2x$ or $\sqrt{4} = \sqrt{e^{2x}}$ seen explicitly
	A1	For $x = \ln 2$
(b)	M1	For differentiating the given expression. This must be correct for this mark. $\frac{dy}{dx} = -2e^{2x}$
	M1	For substituting ln2 into their $\frac{dy}{dx} = -2e^{2\ln 2} = [-8]$
	M1	For a correct method for finding the equation of a straight line using their numerical perpendicular gradient and $y = 0$ and $x = \ln 2$ Award when they substitute into a correct formula, or if they use $y = mx + c$ award when c is obtained. (accept a decimal value for c for this mark awrt $c = -0.087$)
	A1	For $y = \frac{1}{8}x - \frac{1}{8}\ln 2$ or $y = \frac{x}{8} - \frac{\ln 2}{8}$ in exact form only.
(c)	**Area under curve**	
	M1	For a correct statement for the area under the curve with correct limits. Accept the limits either way around. Ignore poor notation. This mark can be implied by later correct work. $\int_0^{\ln 2}\left(4 - e^{2x}\right)dx$
	M1	For a minimally acceptable attempt to integrate as follows. $\int_0^{\ln 2}\left(4 - e^{2x}\right)dx = \left[4x \pm \frac{e^{2x}}{2}\right]$
	M1	For substitution of both limits into their integral of the curve. $A = \left(4\ln 2 - \frac{1}{2}e^{2\ln 2}\right) - \left(0 - \frac{1}{2}e^{0}\right) = \left[4\ln 2 - \frac{3}{2}\right] = [1.27258....]$
	Area of the triangle	
	M1	**Method 1** For a statement of the area. $A = \frac{1}{2}(\ln 2)\left('\frac{1}{8}\ln 2'\right)$ **Method 2** For a statement of the area, with limits either way around. $A = \frac{1}{8}\int_0^{\ln 2}(x - \ln 2)\,dx$ ft their equation of the line.
	M1	For a correct method to evaluate the area of the triangle. **Method 1** $A = \frac{1}{2}(\ln 2)\left('\frac{1}{8}\ln 2'\right) = \frac{1}{16}(\ln 2)^2 = [0.03002.....]$ **Method 2** The integration and substitution must be correct for this mark $A = \frac{1}{8}\int_0^{\ln 2}\lvert x - \ln 2\rvert\,dx = \frac{1}{8}\left\lvert\frac{x^2}{2} - x\ln 2\right\rvert_0^{\ln 2} = \left\lvert\frac{(\ln 2)^2}{16} - \frac{(\ln 2)^2}{8}\right\rvert = \frac{(\ln 2)^2}{16}$

Type MS · Sketching curves and where they cross the axes

Answer 1 of 11 · Question 3 part e · 2 marks

2023 JAN · Paper 02 · extract 19 of 20 · find part e in this mark scheme extract

	Combined area	
	M1	For the total final area using their values $4\ln 2-\frac{3}{2}+\frac{1}{16}(\ln 2)^2$ **This is an A mark in Epen.**
	A1	For awrt 1.3 Also accept the exact Area = $4\ln 2-\frac{3}{2}+\frac{1}{16}(\ln 2)^2$

Type MS · Sketching curves and where they cross the axes

Answer 1 of 11 · Question 3 part e · 2 marks

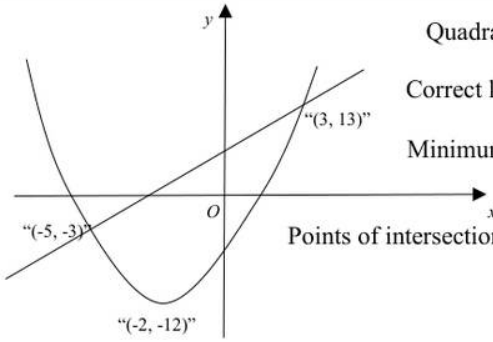
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Type MS · Sketching curves and where they cross the axes

Answer 2 of 11 · Question 2 part c · 4 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

2 (a)	$(x+2)^2-4-8$ $(x+2)^2-12$ $\quad a=2 \quad b=-12$	M1 A1 (2)
(b)	$x^2+4x-8=2x+7$ $x^2+2x-15=0$ $(x-3)(x+5)=0$ or any valid method $x=3, y=13$ $\quad x=-5, y=-3$	M1 dM1 M1 A1 A1 (5)
(c)	 Quadratic drawn Correct line drawn Minimum labelled Points of intersection labelled	B1 B1 B1 ft B1 ft (4) **[11]**

Part	Mark	Additional Guidance
(a)	M1	Use general guidance, allow an expression of the form $\left(x\pm\frac{4}{2}\right)^2\pm q\pm 8 \quad q\neq 0$
	A1	Correct expression as shown, a and b need not be explicitly stated
(b)	M1	Correctly equates the 2 expressions
	dM1	Rearranges to a 3TQ = 0 (allow any 3TQ if intention of rearrangement is clear)
	M1	Uses any valid method to solve – **see general guidance**
	A1	For either pair of values stated
	A1	For all four values, correctly paired or written as coordinates.
	For the final A1 A1, do not allow recovery of y values from part c.	
(c)	B1	Correctly shaped quadratic curve, with a clear minimum point, drawn anywhere on their axis, mark intention.
	B1	Correct line – **must have a positive y intercept, a positive gradient and a negative x intercept**
	B1ft	Correctly labelled coordinates for their minimum, ft their answer from b, **must correctly ft their answer from a, ie minimum point labelled (- *a*, *b*)**
	B1ft	Correctly labelled coordinates for their intersections.
	The coordinates must be clearly indicated and not inferred from a scale on the graph. Ignore any labelling of intersections with axes.	

Type MS · Sketching curves and where they cross the axes

Answer 2 of 11 · Question 2 part c · 4 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sketching curves and where they cross the axes

Answer 3 of 11 · Question 11 part a · 2 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

11 (a)	at P $\quad b=\sqrt{a-2} \Rightarrow b^2=a-2*$	M1A1cso [2]
(b)	At P $y=b \Rightarrow y^2=b^2 \Rightarrow y^2=a-2$	B1
	$V=\pi\int_a^{16}\left(\sqrt{x-2}\right)^2\,\mathrm{d}x-\pi\int_a^{16}(a-2)\,\mathrm{d}x=\pi\int_a^{16}(x-2)\,\mathrm{d}x-\pi\int_a^{16}(a-2)\,\mathrm{d}x$	M1
	$\Rightarrow \pi\int_a^{16}(x-a)\,\mathrm{d}x$ or $\pi\int_a^{16}\left(\sqrt{x-2}\right)^2\,\mathrm{d}x-\pi(a-2)(16-a)$	
	$50\pi=\pi\left[\frac{x^2}{2}-ax\right]_a^{16}$ or $50\pi=\pi\left[\frac{x^2}{2}-2x\right]_a^{16}-\pi(a-2)(16-a)$	depM1A1
	$50\pi=\pi\left[\left(\frac{256}{2}-16a\right)-\left(\frac{a^2}{2}-a^2\right)\right]$	depM1
	or $50\pi=\pi\left[\left(96-\frac{a^2}{2}+2a\right)-\left(18a-a^2-32\right)\right]$	
	$\Rightarrow a^2-32a+156=0$	M1
	$\Rightarrow (a-6)(a-26)=0 \Rightarrow a<16$ so $a=6$	
	$b^2=a-2 \Rightarrow b^2=4 \Rightarrow b=2$	M1A1 A1 [9]
		Total 11 marks

Type MS · Sketching curves and where they cross the axes

Answer 3 of 11 · Question 11 part a · 2 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

Type MS · Sketching curves and where they cross the axes

Answer 3 of 11 · Question 11 part a · 2 marks

2020 JAN · Paper 01 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

Type MS · Sketching curves and where they cross the axes

Answer 3 of 11 · Question 11 part a · 2 marks

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Type MS · Sketching curves and where they cross the axes

Answer 4 of 11 · Question 11 part a · 1 mark

2025 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

11 (a)	$(p=)3e^{\frac{2}{3}}$ oe	B1 [1]
(b)	$\left(\frac{dy}{dx}=\right)e^{\frac{x}{3}} = e^{\frac{2}{3}}$ oe	M1 A1
	$(m_{norm}=)-\frac{1}{"e^{\frac{2}{3}}"}=\left(-e^{-\frac{2}{3}}\right)$ oe	M1
	$y-"3e^{\frac{2}{3}}"="-\frac{1}{"e^{\frac{2}{3}}"}"(x-2)$ oe $\left(y=-e^{-\frac{2}{3}}x+2e^{-\frac{2}{3}}+3e^{\frac{2}{3}}\right)$	M1
	$0-3e^{\frac{2}{3}}=-e^{-\frac{2}{3}}(x-2)$ oe	dM1
	$x=2+3e^{\frac{4}{3}}$	A1
	$(\text{Area of triangle}=)\ \frac{1}{2}\left("3e^{\frac{2}{3}}"\right)\left("2+3e^{\frac{4}{3}}"-2\right)\left(=\frac{9}{2}e^2\right)$	ddM1
	$\int_0^2 3e^{\frac{x}{3}}dx=\left[\frac{3e^{\frac{x}{3}}}{\frac{1}{3}}\right]_0^2$	B1 M1
	$\left(\frac{3e^{\frac{2}{3}}}{\frac{1}{3}}\right)-\frac{3e^0}{\frac{1}{3}}\quad\left(=9e^{\frac{2}{3}}-9\right)$	dM1
	$=\frac{9}{2}e^2+9e^{\frac{2}{3}}-9$	A1cso
ALT for ddM1	$\int_2^{"2+3e^{\frac{4}{3}}"}(-e^{-\frac{2}{3}}x+2e^{-\frac{2}{3}}+3e^{\frac{2}{3}})dx=-e^{-\frac{2}{3}}\frac{x^2}{2}+2e^{-\frac{2}{3}}x+3e^{\frac{2}{3}}x$	
	$=\left[-e^{-\frac{2}{3}}\left(\frac{"2+3e^{\frac{4}{3}}"^2}{2}\right)+2e^{-\frac{2}{3}}\left("2+3e^{\frac{4}{3}}"\right)+3e^{\frac{2}{3}}\left("2+3e^{\frac{4}{3}}"\right)\right]$	ddM1
	$-\left[-e^{-\frac{2}{3}}\left(\frac{"2^2}{2}\right)+2e^{-\frac{2}{3}}(2)+3e^{\frac{2}{3}}(2)\right]$	
		[11]
		Total 12 marks

Type MS · Sketching curves and where they cross the axes

Answer 4 of 11 · Question 11 part a · 1 mark

2025 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	For the correct value of p.
(b)	M1	Differentiates to get an expression of the form $ae^{\frac{x}{3}}$ $a \neq 3$
	A1	For $e^{\frac{2}{3}}$ This must be an exact value, but isw any following decimals.
	M1	Finds the negative reciprocal of their gradient. This is not a completely dependent mark but must come from any attempt at differentiation.
	M1	For a complete unsimplified attempt (if using $y - y_1 = m(x - x_1)$ to find the equation of l, using their value of p and their gradient, which can be any changed gradient after finding an expression and value for $\frac{dy}{dx}$ If $y = mx + c$ is used, a complete attempt must involve correctly rearranging (using their values for p and their gradient) to find c. **Different to question 2, if the candidate is not working with correct values and the rearrangement to find c with exact values is not shown, this mark cannot be awarded**
	dM1	Sets their equation = 0 and attempts to find a value for x. Allow errors in rearrangement. Dependent on previous method mark. This must be an exact value, but isw any following decimals.
	A1	Finds the correct value of x
	ddM1	For correct substitution into the formula for the area of a triangle, using (their x-intercept – 2) and their answer for p Does not need to be simplified. Dependent on previous 2 method marks. Or for stating the integral, with their p and their x intercept as limits and correct integration of at least 2 terms. **We don't need to see substitution of limits for this mark.** Their integral must be of the minimum form $\int_2^{"f+ge^{\frac{h}{3}x}"} \left(ae^{-\frac{c}{3}x} + be^{-\frac{d}{3}} \right) dx$ $a,b,c,d,f,g,h \neq 0$ See notes for next B mark, can be awarded if limits are correct, within a combined integral.
	B1	For stating $\int_0^2 3e^{\frac{x}{3}}$ Note: some candidates are listing this integral by adding or subtracting it from the equation of the line. ie $\int_a^{"b"} \left(-e^{-\frac{2}{3}x} + 2e^{-\frac{2}{3}} + 3e^{\frac{2}{3}} \pm 3e^{\frac{x}{3}} \right) dx$ - mark B1 if limits are correct and B0 if not and potentially next M1 dM1 for anything that is obviously their line $\pm 3e^{\frac{x}{3}}$
	M1	Integrates to get an expression of the form $ae^{\frac{x}{3}}$ $a \neq 3$
	dM1	Substitution of correct limits into any changed expression, must see substitution of $x = 2$ and 0 (unless the 0 limit gives an evaluation of 0) Allow a correct final answer or a correct evaluation of this integral $9e^{\frac{2}{3}} - 9$ to imply this mark, if no errors in working. Examiners are not expected to evaluate to check incorrect expressions integrated. Dependent on previous method mark.
	A1cso	For a full and correct worked solution with no errors.

Type MS · Sketching curves and where they cross the axes

Answer 4 of 11 · Question 11 part a · 1 mark

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Type MS · Sketching curves and where they cross the axes

Answer 5 of 11 · Question 11 part a.i · 0 marks

2023 JAN · Paper 02 · find part a.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Sketching curves and where they cross the axes

Answer 6 of 11 · Question 11 part a.i · 0 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 3 · find part a.i in this mark scheme extract

11 (a)(i)	$(a=)-8$	B1 [1]
(a)(ii)	$e^{2x}=9\Rightarrow 2x=\ln 9$	M1
	$x=\frac{1}{2}\ln 9=\ln 3$ *	A1cso [2]
ALT	$e^{2x}=9\Rightarrow\left[\left(e^{x}\right)^{2}=3^{2}\right]\Rightarrow e^{x}=3$	M1
	$x=\ln 3$ *	A1cso
(b)	$\pi\int_{0}^{\ln 3}\left(e^{2x}-9\right)^{2}dx$	M1
	$\pi\int_{0}^{\ln 3}\left(e^{4x}-18e^{2x}+81\right)dx$	A1
	$(\pi)\left[\frac{1}{4}e^{4x}-9e^{2x}+81x\right]_{(0)}^{("\ln 3")}$ oe	M1 A1ft
	$(\pi)\left[\left(\frac{81}{4}-81+81\ln 3\right)-\left(\frac{1}{4}-9\right)\right]$	M1
	$\pi(81\ln 3-52)$	A1 [6]
		Total 9 marks

Type MS · Sketching curves and where they cross the axes

Answer 6 of 11 · Question 11 part a.i · 0 marks

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π must only be present for the final A1 mark and the first M1 mark (can be implied by first or final A1)

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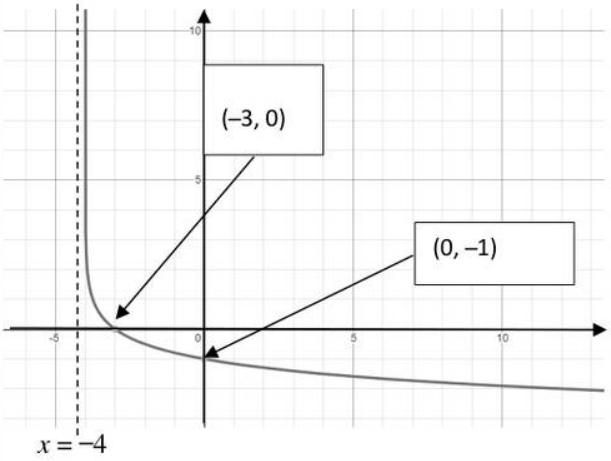
Answer 6 of 11 · Question 11 part a.i · 0 marks

2022 JAN · Paper 01 (regional R) · extract 3 of 3 · find part a.i in this mark scheme extract

Type MS · Sketching curves and where they cross the axes

Answer 7 of 11 · Question 7 part a · 4 marks

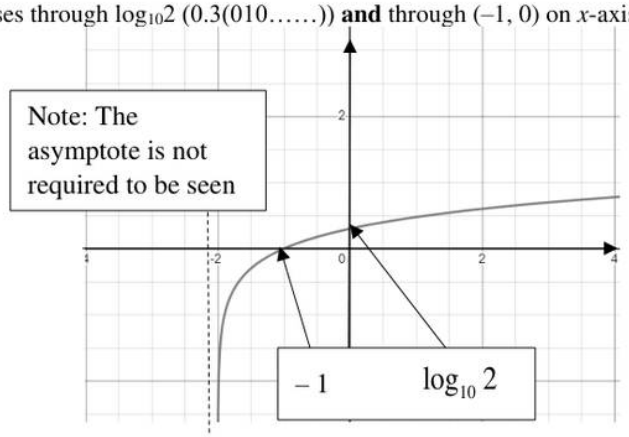
2024 JUN · Paper 01 (regional R) · find part a in this mark scheme extract

7(a)		B1 Shape and position B1 Intersection with x-axis B1 Intersection with y-axis B1 Asymptote [4]
(b)	$\log_{(x+4)} 256 - \log_4(x+4) = 0 \Rightarrow \dfrac{\log_4 256}{\log_4(x+4)} - \log_4(x+4) = 0$ $\Rightarrow 4 - (\log_4(x+4))^2 = 0 \Rightarrow (\log_4(x+4))^2 = 4$ $\Rightarrow \log_4(x+4) = \pm 2$ $x + 4 = 4^2 \Rightarrow x = 12$ $x + 4 = 4^{-2} \Rightarrow x = -\dfrac{63}{16}$ or -3.9375	M1 M1 M1 A1 A1 [5]
		Total 9 marks

Type MS · Sketching curves and where they cross the axes

Answer 8 of 11 · Question 7 part a · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

7 (a)	Shape of the curve, **crossing** negative x-axis and positive y-axis with the asymptotic nature of the curve shown. passes through $\log_{10}2$ (0.3(010……)) **and** through (–1, 0) on x-axis  	B1 B1 [2]
(b)	$\left(2\log_a 4+2\log_a 4^2\right)$ or $(2\log_a 2^2+2\log_a 2^4)$ $2\log_a 4+4\log_a 4$ or $4\log_a 2+8\log_a 2$ $\log_a 4=\frac{1}{6}$ or $\log_a 2=\frac{1}{12}$ oe $a^{\frac{1}{6}}=4$ or $a^{\frac{1}{12}}=2$ $a=4096$ **ALT** $(2)\log_a 64=1\Rightarrow\log_a 64=\frac{1}{2}$ $\Rightarrow a^{\frac{1}{2}}=64$ $\Rightarrow a=4096$	M1 dM1 A1 [M1 dM1 A1] [3]
(c)	$\log_q 16\rightarrow\frac{\log_2 16}{\log_2 q}$ or $\log_2 q\rightarrow\frac{\log_q q}{\log_q 2}$ or $\frac{1}{\log_q 2}$ $\left(5\frac{\log_2 16}{\log_2 q}+4\log_2 q=24\right)$ $\left(5\log_q 16+4\frac{\log_q q}{\log_q 2}=24\right)$ $20+4(\log_2 q)^2=24\log_2 q$ $20(\log_q 2)^2+4=24\log_q 2$ $(\log_2 q-5)(\log_2 q-1)(=0)$ $(5\log_q 2-1)(\log_q 2-1)(=0)$ $\left(\log_2 q=5\right)\rightarrow q=32$ $\left(\log_q 2=\frac{1}{5}\right)\rightarrow q=32$ $\left(\log_2 q=1\right)\rightarrow q=2$ $\left(\log_q 2=1\right)\rightarrow q=2$	M1 M1 A1 dM1 M1 A1 [6]
		Total 11 marks

Type MS · Sketching curves and where they cross the axes

Answer 8 of 11 · Question 7 part a · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	The curve must not bend back on itself and should cross both axes once on negative x and once on positive y. It needs to demonstrate an asymptotic nature. Accept an asymptote drawn correctly as well as their curve.
	B1	It is enough for -1 to be indicated on the x-axis with the curve passing through this point and for $\log_{10}2$ to be marked on the y- axis (decimal 0.3…… allowed) There must be a curve drawn (even incorrectly) to achieve this mark. **Do not** accept intersections unless they are marked on the graph in the correct place.
(b)	M1	For the use of the power rule with logs to obtain $2\log_a 4$ from $\log_a 16$ OR $4\log_a 2$ from $\log_a 16$
	dM1	A **correct** rearrangement of their equation to obtain for example $a^{\frac{1}{6}} = 4$ or $a^{\frac{1}{12}} = 2$ or even of the type $a^1 = 64^2$ There are many ways of completing this. Look for a correct value for the corresponding correct power of a. **This mark is dependent on the first M mark**. Look for correct log work for their values.
	A1	For $a = 4096$
ALT	M1	For use of the correct log rule to obtain $\log_a 64$
	dM1	A correct rearrangement of their equation to obtain $a^{\frac{1}{2}} = 64$ **This mark is dependent on the first M mark**. Look for correct log work for their values.
	A1	For $a = 4096$
(c)	M1	For a fully correct change of base of log to either base 2 or base q
	M1	For an attempt to rearrange and form a 3TQ quadratic. Allow one arithmetical slip only.
	A1	For the correct 3TQ.
	dM1	**Dependent on the second M mark** – an acceptable attempt to solve the quadratic equation – see general guidance.
	M1	For a correct application using either of their solutions moving from log form to exponential form. **Note this is an independent mark.**
	A1	**Both** correct solutions.

Type MS · Sketching curves and where they cross the axes

Answer 8 of 11 · Question 7 part a · 2 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sketching curves and where they cross the axes

Answer 9 of 11 · Question 6 part a · 3 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 6 · find part a in this mark scheme extract

6 (a)	$x^2-5x+4=(x-4)(x-1)$	M1
	$A(1,0)$ $B(4,0)$	A1 A1 (3)
	Notes	
(a)		
M1	For solving the quadratic	
A1	For $A(1,0)$	
A1	For $B(4,0)$	
(b)	$\frac{dy}{dx}=2x-5$	M1
	When $x=1$ $\frac{dy}{dx}=-3$	A1
	Tangents meet on the axis of symmetry of curve C so $x=\frac{1+4}{2}=\frac{5}{2}$	M1 A1
	When $x=\frac{5}{2}$ $y=-3\left(\frac{5}{2}-1\right)=-\frac{9}{2}$	M1
	$\left(\frac{5}{2},-\frac{9}{2}\right)$	A1 (6)
	Notes	
(b)		
M1	For $\frac{dy}{dx}=2x-5$ (Allow if seen in part(c))	
A1	$\frac{dy}{dx}=-3$ when $x=1$	
M1	For use of $\frac{x_1+x_2}{2}$ or $\frac{-b}{2a}$ or $\frac{dy}{dx}=0$	
A1	$x=\frac{5}{2}$	
M1	For substitution of $x=\frac{5}{2}$ into $y-y_1=m(x-x_1)$ oe	
A1	For $\left(\frac{5}{2},-\frac{9}{2}\right)$	

Type MS · Sketching curves and where they cross the axes

Answer 9 of 11 · Question 6 part a · 3 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 6 · find part a in this mark scheme extract

	Alternative Method $\frac{dy}{dx} = 2x - 5$ When $x = 1$ $\frac{dy}{dx} = -3$ So $y = -3x + 3$ When $x = 4$ $\frac{dy}{dx} = 3$ So $y = 3x - 12$ Meet when $-3x + 3 = 3x - 12$ $\left(\frac{5}{2}, -\frac{9}{2}\right)$	M1 A1 M1 A1 M1 A1 (6)
	Notes	
M1	**Alternative** For an attempt to find the equation of the tangent of C at A	
A1	For $y = -3x + 3$	
M1	For an attempt to find the equation of the tangent of C at B	
A1	For $y = 3x - 12$	
M1	For equating the two equations	
A1	For $\left(\frac{5}{2}, -\frac{9}{2}\right)$	

Type MS · Sketching curves and where they cross the axes

Answer 9 of 11 · Question 6 part a · 3 marks

2020 NOV · Paper 02 (regional R) · extract 3 of 6 · find part a in this mark scheme extract

(c)	Gradient of normal at (0, 1) = $\frac{1}{3}$	M1
	When $x=\frac{5}{2}$ $y=\frac{1}{3}\left(\frac{5}{2}-1\right)=\frac{1}{2}$	M1
	$\left(\frac{5}{2},\frac{1}{2}\right)$	A1 (3)
	Alternative Method $y=\frac{1}{3}x-\frac{1}{3}$ and $y=-\frac{1}{3}x+\frac{4}{3}$	M1
	Meet when $\frac{1}{3}x-\frac{1}{3}=-\frac{1}{3}x+\frac{4}{3}$	M1
	$\left(\frac{5}{2},\frac{1}{2}\right)$	A1 (3)

	Notes
(c)	
M1	For Gradient of normal at (0, 1) = $\frac{1}{3}$ ft the gradient found in part (b)
M1	For substitution of $x=\frac{5}{2}$ into $y-y_1=m\left(x-x_1\right)$ oe
A1	For $\left(\frac{5}{2},\frac{1}{2}\right)$
	Alternative
M1	For $y=\frac{1}{3}x-\frac{1}{3}$ and $y=-\frac{1}{3}x+\frac{4}{3}$
M1	For equating the two equations
A1	For $\left(\frac{5}{2},\frac{1}{2}\right)$

Type MS · Sketching curves and where they cross the axes

Answer 9 of 11 · Question 6 part a · 3 marks

2020 NOV · Paper 02 (regional R) · extract 4 of 6 · find part a in this mark scheme extract

(d)	Area $= \frac{1}{2}(4-1)\times\left(\frac{1}{2}+\frac{9}{2}\right)=\frac{15}{2}$	M1 M1 A1 (3)
	Alternative Method 1 Area $= \frac{1}{2}\times 3\times\frac{1}{2}+\frac{1}{2}\times 3\times\frac{9}{2}=\frac{15}{2}$	M1 M1 A1 (3)
	Alternative Method 2 $AN=\sqrt{1.5^2+0.5^2}=\frac{\sqrt{10}}{2}$ $AT=\sqrt{1.5^2+4.5^2}=\frac{3\sqrt{10}}{2}$	M1
	Area $= 2\times\frac{1}{2}\times\frac{\sqrt{10}}{2}\times\frac{3\sqrt{10}}{2}=\frac{30}{4}=\frac{15}{2}$	M1 A1 (3)
	Alternative Method 3 Area $= \frac{1}{2}\begin{vmatrix} 1 & \frac{5}{2} & 4 & \frac{5}{2} & 1 \\ 0 & -\frac{9}{2} & 0 & \frac{1}{2} & 0 \end{vmatrix} \Rightarrow \frac{1}{2}\left(-\frac{9}{2}+2+18-\frac{1}{2}\right)=\frac{15}{2}$	M1 M1 A1 (3)
		[15]

	Notes
(d)	
M1	For $\frac{1}{2}\times AB\times NT$
M1	For $\frac{1}{2}(4-1)\times\left(\frac{1}{2}+\frac{9}{2}\right)$
A1	For $\frac{15}{2}$
	Alternative 1
M1	For area of triangle ANB + area of triangle ATB
M1	For $\frac{1}{2}\times 3\times\frac{1}{2}+\frac{1}{2}\times 3\times\frac{9}{2}$
A1	$\frac{15}{2}$
	Alternative 2
M1	For finding AN and AT
M1	For $2\times\frac{1}{2}\times AN\times AT$
A1	$\frac{15}{2}$

Type MS · Sketching curves and where they cross the axes

Answer 9 of 11 · Question 6 part a · 3 marks

2020 NOV · Paper 02 (regional R) · extract 5 of 6 · find part a in this mark scheme extract

	Alternative 3
M1	Use of area $= \frac{1}{2}\begin{vmatrix} 1 & a & 4 & c & 1 \\ 0 & b & 0 & d & 0 \end{vmatrix}$ oe where (a, b) and (c, d) are the coordinates of T and N
M1	$\frac{1}{2}\left(-\frac{9}{2}+2+18-\frac{1}{2}\right)$
A1	$\frac{15}{2}$

Type MS · Sketching curves and where they cross the axes

Answer 9 of 11 · Question 6 part a · 3 marks

2020 NOV · Paper 02 (regional R) · extract 6 of 6 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

2023 JAN · Paper 02 · extract 1 of 20 · find part c in this mark scheme extract

3 a	$8\left(x^2+\frac{10}{8}x\right)-3$	M1
	$8\left(x+\frac{5}{8}\right)^2-\frac{200}{64}-3=8\left(x+\frac{5}{8}\right)^2-\frac{49}{8}$	M1
	So $A=8$ $B=\frac{5}{8}$ $C=-\frac{49}{8}$	A1 (3)
b (i)	$x=-\frac{5}{8}$	B1 ft
(ii)	$f(x)_{min}=-\frac{49}{8}$	B1 ft (2)
c	$8\left(x+\frac{5}{8}\right)^2-\frac{49}{8}=0\Rightarrow x=-\frac{3}{2}$ $x=\frac{1}{4}$	M1 A1 (2)
d	$8x^2+10x-3=2x+13\Rightarrow 8x^2+8x-16=0$ oe	M1
	$x^2+x-2=0=(x+2)(x-1)\Rightarrow x=-2,\ x=1$	M1 A1
	$y=9$ $y=15$ Coordinates are $(-2,9)$ and $(1,15)$	A1 (4)
e	Correct curve or line Correct curve and line	B1 B1 (2)
		Total 13 marks

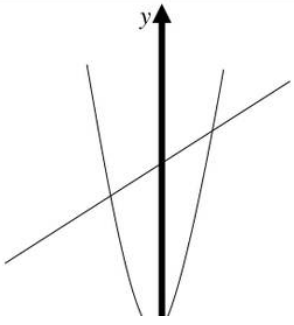
Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

2023 JAN · Paper 02 · extract 2 of 20 · find part c in this mark scheme extract

(b) (i)	B1ft	For $x=-\frac{5}{8}$ ft their $-B$ Allow differentiation: $\frac{dy}{dx}=16x+10=0\Rightarrow x=-\frac{10}{16}$ which must be correct.
(ii)	B1ft	For $f(x)_{min}=-\frac{49}{8}$ ft their C If they differentiate, allow a ft from their differentiation.
	Incorrect or no labelling of parts. If the responses are labelled incorrectly, mark as labelled. If there is no labelling, treat the first as (i) and the second as (ii) and mark accordingly.	
(c)	M1	For setting the given f(x) = 0 and solving using any method. See General Guidance.
	A1	For $x=-\frac{3}{2}$ and $x=\frac{1}{4}$
(d)	M1	For setting $8x^2+10x-3=2x+13$ and forming a 3TQ which as a minimum must be $8x^2+8x\pm C$ where C is a constant, or any simplification, e.g., $x^2+x\pm K$ where K is a constant.
	M1	For attempting to solve their 3TQ. See General Guidance.
	A1	For **both** $x=-2$, $x=1$
	A1	For $(-2,9)$ and $(1,15)$ Accept $x=-2$, $y=9$ and $x=1$, $y=15$ paired correctly.
	ALT – solves the equation in y	
	M1	Substitutes $x=\frac{y-13}{2}$ into $y=8x^2+10x-3$ and forms a 3TQ which as a minimum must be of the form $2y^2-36y\pm X$ $X\neq 0$ $\Rightarrow y=8\left(\frac{y-13}{2}\right)^2+10\left(\frac{y-13}{2}\right)-3\Rightarrow 2y^2-36y+75=0$
	M1	For attempting to solve their 3TQ. See General Guidance.
	A1	For **both** $y=9$ and $y=15$
	A1	For $(-2,9)$ and $(1,15)$ Accept $x=-2$, $y=9$ and $x=1$, $y=15$ paired correctly.
(e)	B1	For correct curve (intersections with x-axis are at $(-1.5, 0)$ and $(0.25, 0)$) **or** line (Intersections are $(-6.5, 0)$ and $(0, 13)$ drawn. **We do not need to see any of these points marked. These are for guidance only.** The question asks for a sketch, accept a reasonable attempt. • Accept a positive quadratic curve with the minimum point below the x-axis, and one branch either side of the y-axis. • Accept a straight line with a positive gradient where the intersection with the y-axis is positive.
	B1	For correct curve **and** line drawn

SKETCH OF CURVE AND LINE



Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

2023 JAN · Paper 02 · extract 3 of 20 · find part c in this mark scheme extract

Question number	Scheme	Marks
4	When $x=\frac{\pi}{2}$ $y=\frac{\pi^3}{8}$ So $\left(\frac{\pi}{2},\frac{\pi^3}{8}\right)$	B1
	$\frac{dy}{dx}=3x^2\sin x+x^3\cos x$	M1 A1 A1
	When $x=\frac{\pi}{2}$ $\frac{dy}{dx}=3\left(\frac{\pi}{2}\right)^2\sin\left(\frac{\pi}{2}\right)+\left(\frac{\pi}{2}\right)^3\cos\left(\frac{\pi}{2}\right)=\left[,\frac{3\pi^2}{4},\right]$	M1
	$y-\frac{\pi^3}{8}=\frac{3\pi^2}{4}\left(x-\frac{\pi}{2}\right)$	M1
	$y=\frac{3}{4}\pi^2x-\frac{1}{4}\pi^3$	A1
		Total 7 marks

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

2023 JAN · Paper 02 · extract 4 of 20 · find part c in this mark scheme extract

Mark	Notes
Note: In this question, all substitution of angle values must be in Radians only.	
B1	For obtaining $y = \frac{\pi^3}{8}$ [allow $\left(\frac{\pi}{2}\right)^3$ and also awrt $y = 3.88$]
M1	For an attempt to use the product rule. The definition of an attempt is as follows: • There must be a correct attempt to differentiate both terms. $\sin x \Rightarrow \cos x \quad x^3 \Rightarrow ax^2$ where $x \neq 0$ • The correct formula must be used. i.e., it must be a sum of their two terms.
A1	At least one term must be correct. Either $3x^2 \sin x$ or $x^3 \cos x$
A1	For $3x^2 \sin x + x^3 \cos x$ oe Ignore any subsequent simplification once you have seen the correct answer – even if the simplification is incorrect.
M1	For substitution of $\frac{\pi}{2}$ into their $\frac{dy}{dx} = \left['\frac{3}{4}\pi^2'\right]$ provided it is a changed expression. Allow a value of awrt 7.4(0) **NOTE:** You must see a full substitution of $\frac{\pi}{2}$ into their $\frac{dy}{dx}$ if their expression for $\frac{dy}{dx}$ is incorrect.
M1	For a correct method for finding the equation of a line using their value of y, dy/dx and the given x [allow $x = 1.57\ldots$]. This must be applied correctly. Either uses the formula with their values, or if uses $y = mx + c$ they must reach a value for c before this mark can be awarded. Do **not** allow processing errors to find the value of c
A1	For $y = \frac{3}{4}\pi^2 x - \frac{1}{4}\pi^3$ Allow $y = 7.4(0)x - 7.75$ or better values. Do not allow a mixture of decimals and exact values.

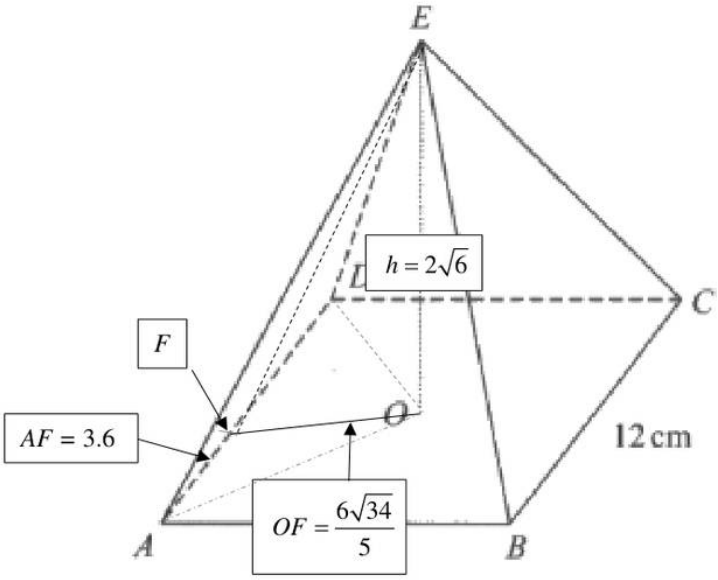
Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

2023 JAN · Paper 02 · extract 5 of 20 · find part c in this mark scheme extract

Question number	Scheme	Marks
5 a	$OC = \frac{1}{2}\sqrt{12^2+12^2} = \left[6\sqrt{2}\right]$ or $AC = \sqrt{12^2+12^2} = \left[12\sqrt{2}\right]$	M1
	$h = EO = 6\sqrt{2} \times \frac{\sqrt{3}}{3} = 2\sqrt{6}$ oe	M1 A1
		(3)
b	Midpoint of AD to $F = \left(6 - \frac{12}{1+4}\right)$ [= 3.6]	M1
	$OF = \sqrt{6^2+3.6^2} = \frac{6\sqrt{34}}{5}$	M1
	$\tan\theta = \frac{2\sqrt{6}}{\frac{6\sqrt{34}}{5}} = 35^\circ$	M1 A1
		(4)
		Total 7 marks

USEFUL SKETCH



Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

2023 JAN · Paper 02 · extract 6 of 20 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For using a correct Pythagoras theorem to find either OC, OA $\left(6\sqrt{2}\right)$ or AC $\left(12\sqrt{2}\right)$
	M1	For using the correct trigonometry (tan ratio): $h = '6\sqrt{2}'\tan 30^\circ = \left[2\sqrt{6}\right]$ or equivalent. Eg $\tan 30 = \dfrac{h}{6\sqrt{2}} \Rightarrow h = ...$
	A1	For $2\sqrt{6}$ oe
(b)	M1	For correct expression for the midpoint of AD to F $\left(6 - \dfrac{12}{1+4}\right)$ or 3.6 oe seen
	M1	For the correct use of Pythagoras' to find OF $\dfrac{6\sqrt{34}}{5}$ oe [6.997..] ft their 3.6
	ALT	
	M1	Finds the length OF using cosine rule. $OF = \sqrt{2.4^2 + \left(6\sqrt{2}\right)^2 - 2\times 2.4\times 6\sqrt{2}\times\cos 45} = \left[\dfrac{6\sqrt{34}}{5}\right]$ OR $OF = \sqrt{9.6^2 + \left(6\sqrt{2}\right)^2 - 2\times 9.6\times 6\sqrt{2}\times\cos 45} = \left[\dfrac{6\sqrt{34}}{5}\right]$ OR $\cos 45^\circ = \dfrac{9.6^2 + \left(6\sqrt{2}\right)^2 - OF^2}{2\times 9.6\times 6\sqrt{2}} \Rightarrow OF = ...$ oe [6.997..]
	M1	For the correct evaluation of their cosine rule. $\left(\dfrac{6\sqrt{34}}{5}\text{ oe}\right)$ Accept awrt 7.00 [6.997142…]
	dM1	For the correct use of $\tan\theta = \dfrac{'EO'}{'OF'}$ This mark is dependent on both previous M marks. They must have a valid method to find OF for the award of this mark.
	A1	For 35(°) or better (Calculator value is 34.997…°.)
	Note: There are other methods – if unsure, send to Review.	

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

2023 JAN · Paper 02 · extract 7 of 20 · find part c in this mark scheme extract

Question number	Scheme	Marks
6 a	$r = \frac{q(2p-3)}{q(2p+3)} = \frac{q(2p+3)}{q(4p+1)}$	M1
	$(4p+1)(2p-3) = (2p+3)^2$	dM1
	$2p^2 - 11p - 6 = 0$	A1
	$(2p+1)(p-6) = 0$	M1
	$p = -\frac{1}{2}$ or $p = 6$	A1 (5)
b	When $p = 6$ $r = \frac{3}{5}$ and $U_1 = q(4\times'6'+1) = (25q)$	M1
	$S_\infty = \frac{25q}{\frac{2}{5}} = 250 \Rightarrow q = 4$	dM1 A1 (3)
		Total 8 marks

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

2023 JAN · Paper 02 · extract 8 of 20 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For $r = \frac{\text{3rd term}}{\text{2nd term}} = \frac{\text{2nd term}}{\text{1st term}}$
	dM1	For attempting to remove the denominators/simplifying the expression to attempt to obtain a 3TQ. Allow one processing error. Allow a minimally acceptable $4p^2 \pm Xp \pm Y = (0)$ or $2p^2 \pm Pp \pm Q = (0)$ $X, Y, P, Q \neq 0$ **Note:** This mark is dependent on the previous M mark.
	A1	For obtaining $2p^2 - 11p - 6 = (0)$ or equivalent. Their working will give them $4p^2 - 22p - 12 = (0)$ before further simplification
	M1	For a valid attempt to solve the 3TQ. Note this is an independent mark for their 3TQ. See General Guidance. The 3TQ must have come from some attempted manipulation involving $q(2p-3)$, $q(2p+3)$ and $q(4p+1)$
	A1	For $p = -\frac{1}{2}$ or $p = 6$
(b)	M1	For finding: • A value of r for their p. Allow any value of r $r \neq 0$ E.g. $r = \frac{(2\times'6'-3)}{(2\times'6'+3)} = ...$ $\left(\text{Note: for } p = -\frac{1}{2} \quad r = -2\right)$ • A value for U_1 $U_1 = q(4\times'6'+1) = ...$ $\left(\text{Note: for } p = -\frac{1}{2}, U_1 = -q\right)$ You may see a = 100 after later working to find the sum to infinity. If their values of p are incorrect, we **must** see working here. **NOTE:** This is a B mark in Epen.
	dM1	For use of $S_\infty = \frac{a}{1-r} = 250$ with their r and their U_1 provided $\lvert r\rvert < 1$ If they use the formula for the sum to infinity on an $\lvert r\rvert > 1$ withhold this mark even if they use it on one valid and one invalid attempt. **Note:** This mark is dependent on the previous M mark.
	A1	For $q = 4$

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

2023 JAN · Paper 02 · extract 9 of 20 · find part c in this mark scheme extract

Question number	Scheme	Marks
7 a	$\frac{dy}{dx} = 2e^{2x}\cos 2x - 2e^{2x}\sin 2x$	M1 A1 A1
	$\frac{dy}{dx} = 2y - 2e^{2x}\sin 2x$ *	A1 cso (4)
b	$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$	M1 A1
	$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} + 2\left(\frac{dy}{dx} - 2y\right) - 4y$	dM1ddM1
	$\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$ *	A1 cso (5)
		Total 9 marks

Part	Mark	Notes
(a)	M1	For an attempt to use the product rule. The definition of an attempt is as follows: • There must be a correct attempt to differentiate both terms. $\cos 2x \Rightarrow \pm 2\sin 2x \quad e^{2x} \Rightarrow 2e^{2x}$ where $x \neq 0$ • The terms must be added The correct formula must be used.
	A1	For one term correct
	A1	For both terms correct
	A1 cso	For obtaining the given result with no errors seen.
(b)	**Method 1**	
	M1	For attempting to differentiate $\frac{dy}{dx}$ to obtain $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 4e^{2x}\cos 2x - 4e^{2x}\sin 2x - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$ For this mark accept either: $\left(4e^{2x}\cos 2x - 4e^{2x}\sin 2x\right)$ **OR** $-4e^{2x}\sin 2x - 4e^{2x}\cos 2x$
	A1	For the correct $\frac{d^2y}{dx^2}$ (This is an M mark in Epen)
	dM1	For simplifying $\frac{d^2y}{dx^2}$ to obtain $\frac{d^2y}{dx^2} = -8e^{2x}\sin 2x$ (A mark in Epen) **This mark is dependent on the first M mark**

	ddM1	For using the substitution $-2e^{2x}\sin 2x = \frac{dy}{dx} - 2y \Rightarrow -8e^{2x}\sin 2x = 4\frac{dy}{dx} - 8y$ **This mark is dependent on BOTH previous M marks.**
	A1cso	For obtaining the given result with no errors seen. $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$
	Method 2	
	M1	For attempting to differentiate $\frac{dy}{dx}$ to obtain $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 2\frac{dy}{dx} - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$ For this mark accept either: $2\frac{dy}{dx}$ or $-4e^{2x}\sin 2x - 4e^{2x}\cos 2x$
	A1	For the correct $\frac{d^2y}{dx^2}$ **This is an M mark in Epen**
	dM1	For substituting $-4e^{2x}\cos 2x \Rightarrow -2y$ **This is an A mark in Epen**
	ddM1	For using the substitution $-4e^{2x}\sin 2x = 2\frac{dy}{dx} - 4y$
	A1	For obtaining the given result with no errors seen. $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$
	Method 3 – Works form LHS and RHS together.	
	M1	For attempting to differentiate $\frac{dy}{dx}$ to obtain $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 4e^{2x}\cos 2x - 4e^{2x}\sin 2x - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$ For this mark accept either: $\left(4e^{2x}\cos 2x - 4e^{2x}\sin 2x\right)$ **OR** $-4e^{2x}\sin 2x - 4e^{2x}\cos 2x$
	A1	For the correct $\frac{d^2y}{dx^2}$ **This is an M mark in Epen**
	dM1	For simplifying $\frac{d^2y}{dx^2}$ to obtain $\frac{d^2y}{dx^2} = -8e^{2x}\sin 2x$ (A mark in Epen) **This mark is dependent on the first M mark**
	ddM1	Multiplies out the given expression in (b) $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y = 4\left(2e^{2x}\cos 2x - 2e^{2x}\sin 2x\right) - 8\left(e^{2x}\cos 2x\right)$ $= 8e^{2x}\cos 2x - 8e^{2x}\sin 2x - 8e^{2x}\cos 2x$ $= -8e^{2x}\sin 2x$

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

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		This mark is dependent on BOTH previous M marks.
	A1	For a conclusion. A simple # sign, 'shown', 'QED' or underlining is sufficient.

Question number	Scheme	Marks
8	$\alpha+\beta=4k\sqrt{2}$ and $\alpha\beta=2k^4-1$	B1 B1
	$(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta\Rightarrow\left(4k\sqrt{2}\right)^2=66+2\left(2k^4-1\right)$	M1 A1
	$k^4-8k^2+16=0$	M1
	$\left(k^2-4\right)^2=0\Rightarrow k=2$	M1 A1
	$(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3$ **or** $\alpha^3+\beta^3=(\alpha+\beta)\left(\alpha^2+\beta^2-\alpha\beta\right)$	M1
	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ **or** $\alpha^3+\beta^3=(\alpha+\beta)(66-\alpha\beta)$	A1
	$\alpha^3+\beta^3=\left(8\sqrt{2}\right)^3-3\times31\times8\sqrt{2}=280\sqrt{2}$ **or** $8\sqrt{2}(66-31)=280\sqrt{2}$	M1
	p=280	A1 (11)
		Total 11 marks

Mark	Notes
B1	For **either** $\alpha+\beta=4k\sqrt{2}$ **or** $\alpha\beta=2k^4-1$
B1	For **both** $\alpha+\beta=4k\sqrt{2}$ **and** $\alpha\beta=2k^4-1$
M1	For the correct algebra on $\alpha^2+\beta^2$ (in any order) **and** substitution of their values of $\alpha\beta$ and $\alpha+\beta$ providing both sum and product are in terms of k. $(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta\Rightarrow\alpha^2+\beta^2=\left('4k\sqrt{2}'\right)^2-2\left('2k^4-1'\right)$
A1	For obtaining $\left(4k\sqrt{2}\right)^2=66+2\left(2k^4-1\right)$ in any order.
M1	For simplifying to form a 3TQ in k^4 i.e., $4k^4-32k^2+64=0$ oe Accept as a minimum $4k^4-32k^2\pm Q=(0)\quad Q\neq0$
M1	For factorising or solving the 3TQ using any valid method. See General Guidance.
A1	For $k=2$ If they also give $k=-2$ withhold this mark.

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

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Method 1	
M1	For expanding $(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3$ or $\alpha^3+\beta^3=(\alpha+\beta)(\alpha^2+\beta^2-\alpha\beta)$
A1	For obtaining $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ or $\alpha^3+\beta^3=(\alpha+\beta)(66-\alpha\beta)$ This must be such that $\alpha\beta$ and $\alpha+\beta$ can be substituted in directly.
M1	For substitution of $\alpha+\beta$ and $\alpha\beta$ for their positive value of k into a **correct** expansion of $\alpha^3+\beta^3$ **NOTE:** If they do not obtain $k=2$, then full substitution of their numerical value for k into $\alpha+\beta$ **and** $\alpha\beta$ must be seen for the award of this mark. For example: $\alpha^3+\beta^3=\left(4\times\text{'their } k\text{'}\sqrt{2}\right)^3-3\times\left(2\times\text{'their } k^4\text{'}-1\right)\times\left(4\times\text{'their } k\text{'}\sqrt{2}\right)$
A1	For $p=280$
Method 2	
M1	Finds the exact value of α and β Solves the equations $\left(\alpha\beta=2k^4-1 \text{ and } \alpha+\beta=4k\sqrt{2}\right)$ simultaneously to give a value for α and β $\alpha=\frac{31}{\beta}\Rightarrow\alpha+\beta=\frac{31}{\beta}+\beta=8\sqrt{2}\Rightarrow\beta^2-8\sqrt{2}\beta+31=0$ $\Rightarrow\beta=\ldots\quad\alpha=\ldots$
A1	For $\alpha=1+4\sqrt{2}\quad\beta=-1+4\sqrt{2}$ OR $\beta=1+4\sqrt{2}\quad\alpha=-1+4\sqrt{2}$
M1	Substitutes these values into $\alpha^3+\beta^3=p\sqrt{2}$ to find a value for p $\left(1+4\sqrt{2}\right)^3+\left(-1+4\sqrt{2}\right)^3=24\sqrt{2}+256\sqrt{2}=280\sqrt{2}\Rightarrow p=\ldots$
A1	$p=280$

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

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Question number	Scheme	Marks
9		
	$\frac{dA}{dt} = 0.45$	B1
	$V = x^3 \Rightarrow \frac{dV}{dx} = 3x^2$	B1
	$A = 6x^2 \Rightarrow \frac{dA}{dx} = 12x$	B1
	$384 = 6x^2 \Rightarrow x = 8$	M1
	$\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dA} \times \frac{dA}{dt}$	M1
	$\frac{dV}{dt} = 3x^2 \times \frac{1}{12x} \times 0.45 \left[= \frac{9x}{80} \right]$ oe	dM1
	When $x = 8$ $\frac{dV}{dt} = 0.9\,\text{cm}^3/\text{s}$	A1
		Total 7 marks

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

2023 JAN · Paper 02 · extract 14 of 20 · find part c in this mark scheme extract

Mark	Notes
B1	For $\frac{dA}{dt} = 0.45$ seen anywhere in their working Accept other letters, for example S for the area $\frac{dS}{dt} = 0.45$
B1	For $\frac{dV}{dx} = 3x^2$ Accept also other letters in place of x such as r for example.
B1	For $\frac{dA}{dx} = 12x$ Accept also other letters in place of x such as r for example.
M1	For setting $384 = 6x^2$ and proceeding to a correct method leading to a value of x Award this mark when they obtain $x^2 = 64 \Rightarrow x = ...$
M1	For a correct expression of the chain rule seen or implied. i.e., $\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dA} \times \frac{dA}{dt}$ They may complete this in two stages. So you may see for example: $\frac{dx}{dt} = \frac{1}{\frac{dA}{dx}} \times \frac{dA}{dt}$ **AND** $\frac{dV}{dt} = \frac{dx}{dt} \times \frac{dV}{dx}$
dM1	For substituting their values into a **correct** chain rule. $\frac{dV}{dt} = 3(8)^2 \times \frac{1}{12(8)} \times 0.45$ This mark is dependent on the previous M mark scored.
A1	For 0.9 (cm^3/s)

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

2023 JAN · Paper 02 · extract 15 of 20 · find part c in this mark scheme extract

Question number	Scheme	Marks
10 a (i)	$\sin 2\theta = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$ *	M1 A1 cso (2)
a (ii)	$\cos 2\theta = \cos^2\theta - \sin^2\theta = \cos^2\theta - (1-\cos^2\theta)$	M1 M1
	$= 2\cos^2\theta - 1$ *	A1 cso (3)
b	$\sin 2\theta - \tan\theta = 2\sin\theta\cos\theta - \dfrac{\sin\theta}{\cos\theta}$	M1
	$\sin\theta\left(2\cos\theta - \dfrac{1}{\cos\theta}\right)$	M1
	$\sin\theta\left(\dfrac{2\cos^2\theta - 1}{\cos\theta}\right) = \tan\theta\cos 2\theta$ *	dM1 A1cso (4)
c	$\tan x = 0$ so $x = 180$	B1
	$\cos 2x = 0$ so $2x = 90, 270, 450, 630$	M1
	$x = 45, 135, 225, 315$	A1 A1 (4)
		Total 13 marks

Part	Mark	Notes
(a)(i)	M1	For the correct use of: $\sin(A+B) = \sin A\cos B + \cos A\sin B \Rightarrow \sin 2\theta = \sin\theta\cos\theta + \sin\theta\cos\theta$ Allow any letter to be used for this mark.
	A1 Cso	For obtaining the given result $\sin 2\theta = 2\sin\theta\cos\theta$ or allow $\sin(\theta+\theta) = 2\sin\theta\cos\theta$
(a)(ii)	M1	For the correct use of: $\cos(A+B) = \cos A\cos B - \sin A\sin B$ $\Rightarrow \cos 2A = \cos^2 A - \sin^2 A$ or $\cos(A+A) = \cos A\cos A - \sin A\sin A$ Allow any letter to be used for this mark.
	M1	For use of $\sin^2 A + \cos^2 A = 1$ and substituting Allow any letter to be used for this mark.
	A1 cso	For obtaining the given result $\cos 2\theta = 2\cos^2\theta - 1$ or allow $\cos(\theta+\theta) = 2\cos^2\theta - 1$
(b)	M1	For correct use of $\sin 2\theta = 2\sin\theta\cos\theta$ **and** $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

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	M1	For factorising the resulting expression to give $\sin\theta\left(2\cos\theta-\frac{1}{\cos\theta}\right)$ **and** using a common denominator of $\cos\theta$ correctly to obtain $\sin\theta\left(\frac{2\cos^2\theta-1}{\cos\theta}\right)$ oe e.g. $\frac{\sin\theta}{\cos\theta}\left(2\cos^2\theta-1\right)$
	dM1	For use of $2\cos^2\theta-1=\cos 2\theta$ **Note:** This mark is dependent on both previous M marks
	A1 cso	For obtaining the given result
	ALT - Allow candidates to work from both sides. For example.	
	M1	For correct use of $\sin 2\theta=2\sin\theta\cos\theta$ **and** $\tan\theta=\frac{\sin\theta}{\cos\theta}$ $\sin 2\theta-\tan\theta=\tan\theta\cos 2\theta$ $\Rightarrow 2\sin\theta\cos\theta-\frac{\sin\theta}{\cos\theta}=\frac{\sin\theta}{\cos\theta}\left(2\cos^2\theta-1\right)$
	M1	For using a common denominator on the LHS to give: $\frac{2\sin\theta\cos^2\theta-\sin\theta}{\cos\theta}=\frac{\sin\theta\left(2\cos^2\theta-1\right)}{\cos\theta}$
	dM1	Multiplies or factorises one side to make LHS = RHS For example; $\frac{2\sin\theta\left(\cos^2\theta-1\right)}{\cos\theta}=\frac{2\sin\theta\left(\cos^2\theta-1\right)}{\cos\theta}$
	A1 cso	Concludes RHS = LHS therefore shown. Note there must be no errors seen. If unsure, send to Review.
(c)	Some candidates are starting again without using the given expression in (b) and are getting their angles from cos x rather than cos $2x$ Allow the use of θ or any other variable throughout this part of the question. As a general principle of marking this part: • B1 – For only 180° seen. If 0° is included, then B0. • M1 – For either $2x=90,(270,450,630)$ **OR** $x=45^\circ$ or 135° or 225° or 315° or any other correct angle in or out of range. • A1 – For any two correct angles apart from 180° • A1 – For all four correct angles, apart form 180°	
	B1	For $x=180$ If $x=0$ included then B0
	M1	For obtaining $2x=90,(270,450,630)$
	A1	For any two correct angles from 45°, 135°, 225° or 315°
	A1	For all four angles from $\cos x$ or $\cos 2x$
	For extra angles within range, penalise the last A mark only. For example; If they have [coming from a correct method] $x=45°, 67.5°, 135°, 225°, 315°$ score M1A1A0 If they have $x=45°, 67.5°, 135°$ then award M1A1A0	

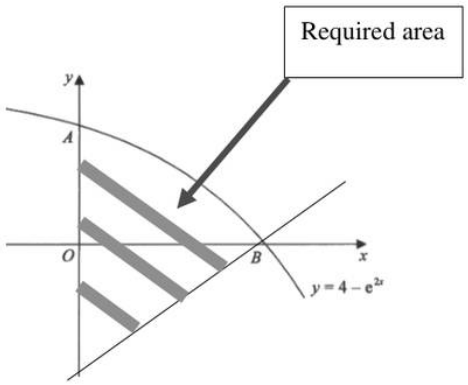
Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

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Question number	Scheme	Marks
11 a	When $x=0 \quad y=3$	B1
	When $y=0 \quad 0=4-e^{2x} \Rightarrow \ln 4 = 2x \Rightarrow x = \ln 2$	M1 A1 (3)
b	$\frac{dy}{dx} = -2e^{2x}$	M1
	When $x=\ln 2 \quad \frac{dy}{dx} = -8$	M1
	$y-0=\frac{1}{8}(x-\ln 2) \Rightarrow y=\frac{1}{8}x-\frac{1}{8}\ln 2$	M1 A1 (4)
c	$\int_0^{\ln 2}\left(4-e^{2x}\right)dx=\left[4x-\frac{1}{2}e^{2x}\right]_0^{\ln 2}=\left(4\ln 2-\frac{1}{2}e^{2\ln 2}\right)-\left(0-\frac{1}{2}e^{0}\right)$	M1M1M1
	$\frac{1}{2}(\ln 2)\left(\frac{1}{8}\ln 2\right)=\frac{1}{16}(\ln 2)^2$	M1M1
	$\frac{1}{8}\int_0^{\ln 2}\lvert x-\ln 2\rvert\,dx=\frac{1}{8}\left\lvert\frac{x^2}{2}-x\ln 2\right\rvert_0^{\ln 2}=\left\lvert\frac{(\ln 2)^2}{16}-\frac{(\ln 2)^2}{8}\right\rvert=\frac{(\ln 2)^2}{16}$	[M1M1]
	$4\ln 2-\frac{3}{2}+\frac{1}{16}(\ln 2)^2=1.3$	M1 A1 (7)
		Total 14 marks

USEFUL SKETCH



Part	Mark	Notes

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

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(a)(i) (ii)	B1	For $y = 3$
	M1	For $\ln 4 = 2x$ or $\sqrt{4} = \sqrt{e^{2x}}$ seen explicitly
	A1	For $x = \ln 2$
(b)	M1	For differentiating the given expression. This must be correct for this mark. $\frac{dy}{dx} = -2e^{2x}$
	M1	For substituting ln2 into their $\frac{dy}{dx} = -2e^{2\ln 2} = [-8]$
	M1	For a correct method for finding the equation of a straight line using their numerical perpendicular gradient and $y = 0$ and $x = \ln 2$ Award when they substitute into a correct formula, or if they use $y = mx + c$ award when c is obtained. (accept a decimal value for c for this mark awrt $c = -0.087$)
	A1	For $y = \frac{1}{8}x - \frac{1}{8}\ln 2$ or $y = \frac{x}{8} - \frac{\ln 2}{8}$ in exact form only.
(c)	**Area under curve**	
	M1	For a correct statement for the area under the curve with correct limits. Accept the limits either way around. Ignore poor notation. This mark can be implied by later correct work. $\int_0^{\ln 2}\left(4 - e^{2x}\right)dx$
	M1	For a minimally acceptable attempt to integrate as follows. $\int_0^{\ln 2}\left(4 - e^{2x}\right)dx = \left[4x \pm \frac{e^{2x}}{2}\right]$
	M1	For substitution of both limits into their integral of the curve. $A = \left(4\ln 2 - \frac{1}{2}e^{2\ln 2}\right) - \left(0 - \frac{1}{2}e^{0}\right) = \left[4\ln 2 - \frac{3}{2}\right] = [1.27258....]$
	Area of the triangle	
	M1	**Method 1** For a statement of the area. $A = \frac{1}{2}(\ln 2)\left('\frac{1}{8}\ln 2'\right)$ **Method 2** For a statement of the area, with limits either way around. $A = \frac{1}{8}\int_0^{\ln 2}(x - \ln 2)\,dx$ ft their equation of the line.
	M1	For a correct method to evaluate the area of the triangle. **Method 1** $A = \frac{1}{2}(\ln 2)\left('\frac{1}{8}\ln 2'\right) = \frac{1}{16}(\ln 2)^2 = [0.03002.....]$ **Method 2** The integration and substitution must be correct for this mark $A = \frac{1}{8}\int_0^{\ln 2}\lvert x - \ln 2\rvert\,dx = \frac{1}{8}\left\lvert\frac{x^2}{2} - x\ln 2\right\rvert_0^{\ln 2} = \left\lvert\frac{(\ln 2)^2}{16} - \frac{(\ln 2)^2}{8}\right\rvert = \frac{(\ln 2)^2}{16}$

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

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	Combined area	
	M1	For the total final area using their values $4\ln 2-\frac{3}{2}+\frac{1}{16}(\ln 2)^2$ **This is an A mark in Epen.**
	A1	For awrt 1.3 Also accept the exact Area = $4\ln 2-\frac{3}{2}+\frac{1}{16}(\ln 2)^2$

Type MS · Sketching curves and where they cross the axes

Answer 10 of 11 · Question 3 part c · 2 marks

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Type MS · Sketching curves and where they cross the axes

Answer 11 of 11 · Question 5 part a · 3 marks

2023 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks
5(a)	$y = Q(x-6)(x+2)$ where Q is a constant	
	Using the coordinates $(4,-6)$	M1
	$-6 = Q(4-6)(4+2) \Rightarrow Q = \frac{-6}{-12} = \frac{1}{2}$	M1
	$y = \frac{1}{2}(x-6)(x+2) \Rightarrow y = \frac{x^2}{2} - 2x - 6$ *	A1 cso
		[3]
(b)	$\frac{dy}{dx} = \frac{2x}{2} - 2$	M1
	$x = 4, \quad \frac{dy}{dx} = 4 - 2 = 2$	M1
	Gradient of normal is $-\frac{1}{2}$	B1ft
	Equation of l: $y - (-6) = -\frac{1}{2}(x-4)$	M1
	$\Rightarrow y = -\frac{x}{2} - 4 \Rightarrow 2y + x + 8 = 0$ *	A1 cso
		[5]
(c)	$\frac{x^2}{2} - 2x - 6 = -\frac{x}{2} - 4 \Rightarrow \frac{x^2}{2} - \frac{3}{2}x - 2 = 0$	M1
	$\Rightarrow (x-4)(x+1) = 0 \Rightarrow x = -1, 4$	M1 A1
	Area $= \int_{-1}^{4}\left(\frac{x^2}{2} - 2x - 6\right) dx - \int_{-1}^{4}\left(-\frac{x}{2} - 4\right) dx$	M1
	$= \left[\frac{x^3}{6} - \frac{3}{4}x^2 - 2x\right]_{-1}^{4}$	M1
	$= \left(\frac{4^3}{6} - \frac{3}{4}\times 4^2 - 2\times 4\right) - \left(\frac{(-1)^3}{6} - \frac{3}{4}\times(-1)^2 - 2\times(-1)\right)$	M1
	$= \left(\frac{4^3}{6} - \frac{3}{4}\times 4^2 - 2\times 4\right) - \left(\frac{(-1)^3}{6} - \frac{3}{4}\times(-1)^2 - 2\times(-1)\right)$	
	$= -\frac{125}{12} \Rightarrow$ Area $= \frac{125}{12}$ $\left(\text{units}^2\right)$ oe	A1
		[7]
		Total 15 marks

Part	Mark	Notes
(a)	M1	Uses the intersections with the x-axes to form a quadratic equation of the form $y = Q(x \pm 6)(x \pm 2)$

Type MS · Sketching curves and where they cross the axes

Answer 11 of 11 · Question 5 part a · 3 marks

2023 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

	M1	Uses their Quadratic with the coordinates $(4, -6)$ to find the value of Q Allow just one processing error here.
	A1 cso	For the correct equation in the required form. Note this equation is given to candidates. Both above steps must be complete and correct for the award of this mark.
	ALT – Uses simultaneous equations	
	M1	Sets up all three equations with the given coordinates. These must be correct. $y = px^2 + qx + r$ $0 = 4p - 2q + r$ **1** $0 = 36p + 6q + r$ **2** $-6 = 16p + 4q + r$ **3**
	M1	Attempts to solve their three simultaneous equations to find the values of p, q and r At least one correct value is evidence of correct method. **2 - 1** $0 = 32p + 8q$ **4** **3 - 2** $6 = 20p + 2q$ **5** **5** $\times 4$ $24 = 80p + 8q$ **6** **6 - 4** $24 = 48p \Rightarrow p = \frac{1}{2}$, $24 = 40 + 8q \Rightarrow q = -2$ $0 = 2 + 4 + r \Rightarrow r = -6$
	A1 cso	For the correct equation in the required form. Note this equation is given to candidates. All of the above steps must be complete and correct for the award of this mark.
(b)	M1	For differentiating the given expression. This must be correct, simplified or unsimplified for this mark.
	M1	For substituting $x = 4$ into their $\frac{dy}{dx}$ to find the gradient of the tangent.
	B1ft	For finding the gradient of the normal. Ft their gradient.
	M1	For forming an equation for l using the equation of the normal which must have come from use of calculus. If they use $y = mx + c$ then they must find c and form an equation for the award of this mark. For example; $c = -4 \Rightarrow y = -\frac{x}{2} - 4$
	A1 cso	For the correct equation in the required form. Accept the terms in any order. For example: even $0 = -8 - x - 2y$
(c)	M1	For equating the equation of S to their l and forming a 3TQ
	M1	For attempting to solve the 3TQ [see General Guidance for the definition of an attempt] to find two points of intersection.
	A1	For **both** correct values of x
	M1	For a correct statement for the area using **their two points of intersection** correctly. **Do not accept limits of $x = -2$ and 4 or 6** They may complete these two areas separately and combine at the end. Check to the end of their work before you score this mark. Accept either $\int_{-1}^{4} \text{Curve} - \int_{-1}^{4} \text{Line}$ or $\int_{-1}^{4} \text{Line} - \int_{-1}^{4} \text{Curve}$ for this mark.
	M1	For an attempt to integrate either the expression for the line or the curve.

Type MS · Sketching curves and where they cross the axes

Answer 11 of 11 · Question 5 part a · 3 marks

2023 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

		See General guidance – but no power of x is to decrease.
	M1	For substituting in their values correctly **AND** subtract the two integrals. Explicit substitution must be seen if the final area is incorrect, or the limits are incorrect. A final correct area which follows correct integration is adequate evidence.
	A1	For the correct area

Question	Scheme	Marks